

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713
NETWORKS

COURSE NAME: COMPUTER

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I R.HARI KRISHNAN (192565075) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R.hari krishnan

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full

notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps)**, **home Wi-Fi (20 Mbps)**, and **mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

COMPUTER NETWORKS

ASSESSMENT TOOL 1

MODEL ANSWERS

Question 1: CRC (Cyclic Redundancy Check)

CRC is an error detection technique used in data communication. The sender treats the binary message as a polynomial and divides it by a generator polynomial using modulo-2 division. The remainder obtained is appended to the original message before transmission. At the receiver side, the same division is performed. If the remainder is zero, the frame is accepted; otherwise it is rejected.

Given:

Original message = 11100011

Generator polynomial = $x^3 + 1 = 1001$

Degree of polynomial = 3

Step 1: Append three zeros to the original message because the polynomial degree is 3.

Dividend = 11100011000

Step 2: Perform modulo-2 (XOR) long division using divisor 1001. The final CRC remainder obtained is 111.

Step 3: Append the remainder to the original message.

Transmitted Codeword = 11100011 + 111 = 11100011111

Error Case:

If the leftmost bit changes during transmission, the receiver gets 01100011111. The receiver again divides this codeword by 1001. Since the remainder is non-zero, the receiver concludes that the received frame has been corrupted during transmission and discards the frame or requests retransmission depending on the protocol.

Conclusion: CRC provides excellent error detection capability with very little overhead and is widely used in Ethernet, Wi-Fi, storage devices and communication protocols.

Question 2(a): Throughput

Bandwidth is the maximum data carrying capacity of a network, whereas throughput is the actual amount of data successfully delivered in a unit time.

Given:

Bandwidth = 10 Mbps

Frames transmitted = 12,000 frames/minute

Frame size = 10,000 bits

Calculation:

Total bits transmitted per minute

= $12,000 \times 10,000$

= 120,000,000 bits/minute

Throughput

= $120,000,000 / 60$

= 2,000,000 bits/second

= 2 Mbps

Interpretation:

Although the available bandwidth is 10 Mbps, the network effectively transfers only 2 Mbps. This indicates that only 20% of the available bandwidth is utilized because of protocol overhead, network delay, congestion, acknowledgments or idle time.

Question 2(b): Shannon Channel Capacity

Shannon's theorem gives the theoretical maximum data rate for a noisy channel.

Formula:

$$C = B \log_2(1 + S/N)$$

Given:

Bandwidth (B) = 3000 Hz

Signal-to-noise ratio = 3162

Calculation:

$$C = 3000 \times \log_2(3163)$$

$$\log_2(3163) \approx 11.63$$

$$C = 3000 \times 11.63$$

$$= 34,890 \text{ bps}$$

$$\approx 34.9 \text{ kbps}$$

Interpretation:

The channel cannot transmit data faster than approximately 34.9 kbps without introducing errors, regardless of the modulation technique used.

Question 3: Hamming (12,8) Error Correcting Code

Hamming code is a forward error correction technique capable of correcting one-bit errors and detecting two-bit errors without retransmission.

Working:

1. Divide the data into 8 information bits.
2. Reserve parity bits at positions 1, 2, 4 and 8.
3. Calculate parity bits using even parity.
4. Transmit the 12-bit codeword.
5. At the receiver, recompute parity bits.
6. The syndrome value identifies the position of the erroneous bit.
7. Flip the erroneous bit to recover the original data.

Application to Hospital Scenario:

Wireless patient monitoring devices may experience electromagnetic interference causing a single-bit error. Hamming (12,8) allows the monitoring station to identify and correct the error immediately without waiting for retransmission. This ensures that doctors receive accurate heart rate and blood pressure values with minimum delay, improving patient safety and reliability.

Question 4(a): IPv4 Subnetting

Given:

Network = 172.16.10.0/24

New subnet mask = /27 = 255.255.255.224

Borrowed bits = 3

Number of subnets = $2^3 = 8$

Host bits = 5

Usable hosts per subnet = ~~2~~ - 2 = 30

Subnet	Network	Host Range	Broadcast
1	172.16.10.172.16.10.0	172.16.10.1-30	172.16.10.31
2	172.16.10.32	172.16.10.33-62	172.16.10.63
3	172.16.10.64	172.16.10.65-94	172.16.10.95
4	172.16.10.96	172.16.10.97-126	172.16.10.127
5	172.16.10.128	172.16.10.129-158	172.16.10.159
6	172.16.10.160	172.16.10.161-190	172.16.10.191
7	172.16.10.192	172.16.10.193-222	172.16.10.223
8	172.16.10.224	172.16.10.225-254	172.16.10.255

IP address 172.16.10.78 lies between 64 and 95, therefore it belongs to subnet 172.16.10.64/27.

Network address = 172.16.10.64

Broadcast address = 172.16.10.95

Usable hosts = 172.16.10.65 to 172.16.10.94

IP address 172.16.10.95 is the broadcast address and cannot be assigned to a host.

Question 4(b): IPv6

Original Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed form:

2001:db8::ff00:42:8329

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

With prefix /64, the network prefix is:

2001:db8:0:0::/64

Remaining host bits = 64

Total possible host addresses = ~~2~~ = 18,446,744,073,709,551,616

IPv6 provides a much larger address space than IPv4, eliminates address exhaustion and supports efficient routing, auto-configuration and improved security.

Question 5: Flow Control Case Study

Primary Networking Problem:

The network suffers from flow control mismatch between the high-speed sender and slower receivers.

Analysis:

The university server transmits encrypted question papers over a 10 Gbps connection. However, students connect using different bandwidths ranging from 5 Mbps mobile networks to 100 Mbps fiber connections. Because the sender transmits data much faster than some receivers can process it, packets accumulate in receiver buffers. Once the buffers become full, packets are dropped. TCP interprets packet loss as congestion, triggering retransmissions and reducing overall efficiency. Delayed acknowledgments further slow communication. As more retransmissions occur, network congestion increases, causing freezing, incomplete downloads and delayed access to examination papers.

Effects:

- Receiver buffer overflow
- Packet loss
- Retransmissions
- Increased latency
- Reduced throughput
- Congestion
- Poor user experience

Solutions:

1. Implement TCP Sliding Window flow control.
2. Use dynamic receive window advertisements.
3. Apply TCP congestion control algorithms.
4. Use Content Delivery Networks (CDNs) or regional servers.
5. Increase receiver buffer size.
6. Balance traffic using load balancers.
7. Compress encrypted files before transmission.

Conclusion:

Flow control ensures that the sender never transmits faster than the receiver can accept. Proper flow control improves reliability, prevents packet loss, minimizes retransmissions and guarantees timely delivery of examination papers to all students.